Horizon-AI 2026 Proposal: NeuroFlow: High-Efficiency Low-Power Edge Neural Acceleration

Proposal ID: PROP-03-FIGURE-CONTRADICTION | **PI:** Dr. Sarah Chen
Consortium: ETH Zurich (Switzerland), IMEC (Belgium), Univ Bologna (Italy)
Requested Grant: €2,150,000.00

Abstract

NeuroFlow introduces an asynchronous event-driven neuromorphic kernel that delivers a +45% throughput acceleration on edge IoT benchmarks over ARM baseline.

Scientific Methodology

We map sparse spike neural networks to a custom asynchronous FPGA substrate. As shown in Figure 2 ('Throughput Speedup across 10 benchmarks'), the measured throughput gains average approximately 11.8% over standard GPU quantization.

Impact And Dissemination

Enables ultra-low power autonomous drone navigation and robotic perception.

Work Packages

WP1: Neuromorphic kernel. WP2: Silicon tape-out. WP3: Edge drone flight trials.

Budget

Compliant personnel €1.10M, clean 25% indirect costs, 7% subcontracting.